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BSE4104 Emerging Trends in Software Engineering
AI Agentic Systems Capstone

University Student-Support Case Agent
Final Engineering Report

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1. Executive Summary

New Makerere University students need a reliable way to complete the first onboarding steps: webmail, campus Wi-Fi, the ACMIS portal, and registration. The public instructions exist, but they are spread across several documents. A general chat model can sound confident while inventing a step that is not in those documents.

The group built a bounded support agent for freshman onboarding only. It answers from an approved corpus, cites the source, and, when the student agrees, saves one support draft for a staff queue. It does not decide admissions, grades, discipline, or fees.

Item	Choice in this release
Answer model	gemini-3.6-flash
Retrieval model	gemini-embedding-001
Tools	retrieve_documents and create_support_case
Who runs a tool	The application, not the model
Limits	Three rounds, three tool calls, one draft per session
Week 3 retrieval	11 of 15 questions (73%)
Week 7 scenarios	29 of 36 (80.5%)
Week 8	Five defects from that evaluation were fixed
Principal limit	Some relevant fee and joining passages still score below 0.70, so the agent refuses rather than guess

2. Problem Context and Requirements

2.1 User and workflow

The user is a freshman who is trying to activate an account or follow a published onboarding step. The workflow is short.

- The student asks a question.
- The agent searches the approved documents.
- If a passage is relevant, the answer is grounded in it and the source title is shown.
- If nothing relevant is found, the agent says so and may offer a support draft.
- The student must click "I agree to save a support draft" before a draft can be written.
- Staff still own the case. The agent cannot resolve, close, or reassign it.

2.2 Requirements and what counts as success

The requirements are recorded in the Agent Task Contract, the user stories and acceptance criteria, and the prompt specification.

Outcome	What it means
Success	A grounded answer with a source, a correct refusal outside freshman onboarding, or a validated draft in the right queue
Wrong success	An invented procedure, a fee or grade decision, a draft saved without agreement, or a second draft in the same session

2.3 AI boundary

- The model may draft language and choose a tool.
- It may not change a university record, approve a refund, alter a grade, or speak for admissions or a disciplinary body.
- High-impact actions stay with a person.
- The corpus is public university guidance plus the group's controlled copies.
- The agent does not connect to ACMIS, the fees ledger, or any live student database.

2.4 Acceptance checks

Acceptance is checked against behaviour, not against a pleasant tone. These checks are the acceptance criteria the Week 7 scenarios and the deterministic script both use.

Case	Pass	Fail
Webmail question	The reply follows a retrieved passage and names the source	The reply invents a step
Continuing student	The reply refuses and does not search	The agent treats the student as a freshman
Support draft	Sidebar agreement is on, the summary is a real sentence, the registration number matches the freshman pattern, and the stored row has a queue	Any of those is missing, a second draft is written in the same session, or the summary asks for a waiver or a refund

3. System Architecture

3.1 Application

The running application is a Streamlit page, src/app.py. The student picks a prompt version in the sidebar. For v1.3 the page calls run_tool_turn in src/tools/orchestrate.py.

Prompt	What the page does
v1.0 and v1.1	Call the model with no documents
v1.2	Adds retrieval
v1.3	The agent. This is the release path

Session data lives in SessionState:

- Session id, conversation, and agreement flag.
- Active case id, retrieved context, iteration count, and the tool log.

3.2 Checks before the model

Deterministic code surrounds the model. Before any model call, Sense checks the current student message. Only a message that passes is sent to Gemini, together with the tool declarations.

Check	Stop
Instruction-override attempt	S6
The student says they are not a freshman	S4
Blocked decision topic	S5

- If the model requests a tool, handlers.py runs retrieve_documents or create_support_case.

- The model never writes the draft file itself.

3.3 Where data lives

Store	Path	Committed
Chunks and embeddings	data/chunks/	Chunks yes. The embedding file no
Drafts	data/cases/	No
One JSON log per v1.3 turn	logs/	No. Samples are kept in evidence/traces/

- The external dependency is the Gemini API, reached with a key in .env.
- There is no other network tool, browser, or payment integration.

3.4 What can be tested without a live model

Sense, validation, and the draft store are ordinary Python. The deterministic suite substitutes a fake Gemini client and a temporary draft file. That is how C07, C08, R04, and the limit checks can fail or pass on a machine with no API key.

The model boundary is also where a 503 stops. The orchestration layer catches it, writes the raw text into the turn log, and returns a short sentence to the page. The student never has to interpret an HTTP status.

4. Foundation Model and Prompt Engineering

4.1 Why this model

The group selected Gemini because the course work needed both a chat model and an embedding model from one provider, with a free-tier key that every member could use. The choice is an engineering trade-off, not a claim that this model is the best general assistant. It is small enough to call from a student project, and the application treats its output as untrusted until the tools and checks have run.

Role	Model
Answers	gemini-3.6-flash
Embeddings	gemini-embedding-001

4.2 Prompt versions

Prompts are files, not strings hidden in the interface. load_prompt reads the version the sidebar selects. The Week 7 and Week 8 release run v1.3.

File	What changed
prompts/v1.0_baseline.txt	Unconstrained baseline
v1.1	Limits the role to freshman onboarding
v1.2	Requires answers from retrieved passages, and a refusal when retrieval is empty
v1.3	Adds the two tools, the agreement rule, and the draft rules

4.3 The important iteration

The important iteration was to stop asking the model to be the safety system.

- Early prompts said the assistant must not decide fees or grades. That sentence is still in the prompt.
- The release also blocks those topics in code before the model is called.
- Fee-decision language inside create_support_case is rejected no matter which category the model picked.

Prompt text is guidance. The stop conditions are the control.

5. Context Engineering and RAG

5.1 Corpus and pipeline

The corpus is eight public Makerere freshman-onboarding documents, listed in the corpus source register. The live timetable site is excluded. It is not a stable document.

Step	Script
Download	download_sources.py
Extract	extract_text.py
Chunk	chunk_text.py
Embed	embed_chunks.py

- Chunks are about 1,500 characters. The Week 3 index held 90 chunks.
- chunks.json is committed so the text can be inspected.
- The embedding file is not committed, because it is large and can be rebuilt.

5.2 Retrieval rule

retrieve.py ranks chunks by cosine similarity and returns at most three chunks at or above 0.70. The threshold was chosen after comparing 0.50, 0.65, and 0.70 on the Week 3 questions. A lower bar returned more passages and also more weak matches. The agent would rather say that it did not find a source than quote a weak one.

5.3 Week 3 result and the misses

Set	Result
All 15 questions	11 of 15 (73%)
Answerable	4 of 5
Unanswerable	5 of 5. The refusal path worked when the corpus did not contain the answer

The misses were mostly threshold misses:

- Joining instructions.
- Eduroam troubleshooting.
- Webmail password reset.
- A combined Wi-Fi question.

Week 7 scenario I02 is the same class of miss. A question about whether functional fees are refundable did not clear 0.70, so the agent refused. Week 8 did not lower the threshold and did not add a keyword index. Either change would need the 15-question set, and the full scenario set, to be scored again.

5.4 How a reply is grounded

Retrieval status	Required reply
ok	Follow the passages and name the source
no_match	Do not fill the gap from general knowledge about universities. Stop S3

The student can still be offered a draft, and that draft records the question. It does not record an answer the documents did not support. This is why a retrieval miss is classified as a remaining limit rather than as a silent success. The student is told the document was not found. A staff member can still answer the fee question from the policy itself.

6. Tools and Agentic Workflow

6.1 Tool contracts

Tool	Input	What it returns or refuses
retrieve_documents	A query	Status, a count, and source titles with similarity. It does not return the whole corpus
create_support_case	Category, issue summary, and the registration number when one was given	A local draft, or a validation error

create_support_case accepts a draft only when all of these hold:

- The registration number matches the freshman pattern, when a draft is tied to a person.
- The summary is not empty and is not a placeholder.
- The summary does not ask to waive, refund, or approve a fee.
- A valid draft is stored locally with an id of the form CASE-year-six digits.

6.2 The loop and the stops

Step	Who acts
Sense	The code checks in section 3.2
Plan	The model, which may request a tool
Act	The handler
Observe	The tool result is fed back for another round
Stop	The turn ends with one label from the table below

Stop	Meaning
S1	Grounded answer
S2	Saved draft, and the hand-off to staff
S3	No matching document, with an offer to draft a case
S4	Out of freshman scope
S5	Blocked topic
S6	Instruction-override attempt
S7	Three-round limit
S8	Tool error after one retry

6.3 Limits and the agreement control

Limits are constants in orchestrate.py.

Limit	Value	Why
Rounds	3	Stops an open-ended loop. This is stop S7
Tool calls per turn	3	Caps how much one turn can do
Case calls per turn	2	One validation error can be corrected
Successful drafts per session	1	A second create for the same registration number is refused by the case history check

- The agreement control is the sidebar button.
- Typing "I agree" in the chat is not enough, because the flag is set only by that button.
- Week 5 traces T1, T2, and T3 show a grounded answer, a draft, and a failure recovery under these rules.

6.4 Registration number and queues

A registration number is accepted only when it matches the freshman form used in the case store: two digits, a slash, U or X, a slash, four or five digits, and an optional /PS or /EVE suffix. The number is read from the student's own messages when the draft is tied to a person. The model is not allowed to invent one.

Queues are a fixed map, not a free-text field. The queue is a label for a person. It is not a dispatch into a live helpdesk system.

Category	Queue
Webmail and Wi-Fi	ICT helpdesk
Registration	Registration office
Fees information	Fees information
Accommodation	Accommodation office
Anything else	General onboarding

7. Memory, State and Interoperability

7.1 Two stores, kept apart

Store	How long it lasts	What a reload does
Session state	While the Streamlit session lasts	Clears the conversation, the agreement flag, and the active case id
Draft file on disk	Until a person removes the local file	Does not clear

That split is intentional. The chat is temporary. The staff draft is the record.

7.2 Case History Recall

Persistent memory is Case History Recall, not a vector memory of the chat.

- If the current student message asks about a prior case, the registration number is taken only from the student's own messages, not from the assistant's text.
- get_case_history reads drafts.json by that exact identifier and injects a note for the rest of the turn.

- The group did not implement a 90-day deletion job. Drafts stay until a person removes the local file. The memory note records that limit.
- Drafts are gitignored so a registration number is not pushed to GitHub.

7.3 Interface, not a live server

The interoperability artefact is an MCP-style interface specification for create_support_case, not a live Model Context Protocol server. The course reading did not cover MCP in the supplied editions, and the assignment allowed a clear interface design.

- The specification states the inputs, the validation errors, the queues, and the actions the tool will not perform.
- A later team could wrap the same handler as an MCP tool without changing the safety checks.

8. Evaluation, Observability and Guardrails

8.1 What was measured

- Week 3 measured retrieval and grounded answers on 15 questions: answerable, partly answerable, and unanswerable.
- Week 7 measured the full agent on 36 scenarios, above the 30-scenario minimum.

A scenario passed only when every applicable criterion passed: task completion, groundedness, tool choice, scope, latency under a 10-second target, and safety. The set mixes live Gemini calls and deterministic checks that use a fake client, so the safety paths do not depend on the model having a good day.

8.2 Results

Group	Result	Why, where it was not a full pass
Week 3	11/15 (73%)	See section 5.3
Week 7	29/36 (80.5%)	The headline is not rewritten after Week 8
Normal grounded questions	Passed	Full pass
Adversarial injection	Passed	Full pass
Limit checks	Passed	Includes the three-round stop
Incorrect information	80%	I02 missed the fee passage
Blocked topic	75%	Two information questions were over-refused
Edge cases	50%	A second-year student was treated as a freshman, and a raw API error was shown in the chat

Latency sometimes exceeded 10 seconds: 13.6 seconds on a memory lookup and 12.5 seconds on case creation. Those turns still completed. The 10-second figure is a target, not a hard stop.

8.3 Guardrails and what a log contains

The guardrails are:

- The Sense stops, the tool allow-list, and schema validation.
- The agreement flag and the draft caps.

- Resolve, close, and reassign are not registered tools.
- Prompt injection is refused before tools run.
- A caller cannot supply a different session id and have it trusted.
- Fee-decision wording is rejected in the summary.

The observability evidence and the guardrail evidence PDFs record these checks. Each v1.3 turn writes one log.

- Turn id, session id, latency, model, round count, and prompt id.
- Whether a case-history note was used.
- The student message, retrieval status, and source titles.
- Tool names, shortened results, any error, and the stop condition.
- A reply preview of at most 400 characters.

Passage bodies and the full prompt are not stored.

8.4 Week 8 remediation and retest

Five failures were code defects and were fixed on tag v1.5.0.

- Information questions about the Code of Conduct or an admission letter are no longer blocked. A request for admissions help or a grade change still stops at S5.
- A 503 or 429 is shown as "The assistant is temporarily busy. Please try again in a few minutes." The raw error remains in the log. If a draft was already saved, the reply still includes the case id and stops at S2.
- Second-year, third-year, fourth-year, fifth-year, and 2nd-year through 5th-year phrasing now stop at S4.
- Placeholder summaries, including "Unspecified onboarding issue", and summaries shorter than 8 characters, are validation errors.
- Fee-decision language is rejected in every category.

Scripted retest, from src/, with no API key:

Command	Result
python run_deterministic_checks.py	16 of 16. C07, C08, R04, and R05 had encoded the expected behaviour and were failing before the fix
python week8_retest.py	5 of 5. The Code of Conduct question and the admission-letter question are not blocked. An admissions decision, a grade change, and the second-year sentence still stop before the model. Recorded in docs/evaluation/week8_retest.json

On 24 October 2026 the 19 live chats were run again from docs/evaluation/week8_runner_input.xlsx. The committed Week 7 results file has 35 scenarios, not the 36 named in the Week 7 write-up. The 29/36 figure is not rewritten. Eighteen chats finished. Replies are in docs/evaluation/week8_live_results.csv.

Scenario	24 October result
B07	Cites the Joining Instructions and lists the registration documents. Stop S1
N03	Stops at S4 with no model call

Scenario	24 October result
I02	Still returns no_match and stops at S3
B06	No longer blocked in code, but the model refuses it and does not search, so the grounded-answer criterion still fails
M01	The memory question stopped because Gemini returned 429 after the free-tier quota was used up. Not scored as a product failure
C05	Saved one fees draft, CASE-2026-000001, in 11.4 seconds

The checked-in traces are the inspection path when a marker cannot run Gemini.

Trace	What it shows
T1	A grounded answer
T2	A draft
T3	A failure that recovers inside the limits
O1	The admissions refusal at S5, with no model call
O2	An instruction-override refusal at S6
O3	A grounded webmail retrieval. During that run the model returned 503, so O3 records the retrieval result and says the model was unavailable
O4	Keeps the live 503, including a latency of 8334 milliseconds, in the log rather than in the chat

Together they show a success, a hand-off, a blocked topic, an attack refused before tools, and a dependency failure that the application can name.

9. Software Engineering Quality and Deployment

9.1 Repository, tags, and how to run

The repository is https://github.com/SumayyaN/BSE41014_capstone_group_m_eve. History is on main through the Week 7 seal. A marker should check out v1.5.0.

Tag	What it is
v1.3.0	The Week 7 snapshot
v1.4.0	The README release before these defect fixes
v1.5.0	The final release, an annotated tag on branch week8-release

Run instructions are in the README, in this order:

- Python 3.11 or newer.
- pip install -r requirements.txt.
- A GEMINI_API_KEY in .env, copied from .env.example.
- The four RAG scripts.
- python -m streamlit run src/app.py from the repository root.

9.2 Secrets, tests, and deployment

- Secrets stay out of git. .env is ignored. The sample file contains the variable name only.
- Drafts, embeddings, raw downloads, and logs are ignored.
- The checked-in traces are the Week 5 T1-T3 samples and the Week 7 O1-O4 samples.

- There is no continuous-integration workflow in the repository.
- The tests that travel with the release are the deterministic script and the Week 8 retest script. They do not need an API key. The live scenario runner does.
- The group did not deploy a public server. The reproducible run is the local Streamlit application from the tag. That matches the assignment option of a deployed or reproducibly runnable release.

9.3 Review

Code review happened through pull requests on the shared repository, including the bounded-agent workflow, the memory work, and the observability logs. Week 8 fixes are small and sit next to the tests that name the old failures. A later maintainer can change a queue or a stop message without rewriting the model prompt, because those rules are tables and constants.

9.4 Reproducibility

Reproducibility was treated as a deliverable, not as a hope. The README names tag v1.5.0, the Python version, the install command, the order of the four corpus scripts, and the Streamlit command. chunks.json is in git so the text of the corpus can be read before embeddings are rebuilt. The embedding file, raw downloads, drafts, and logs are generated on the machine that runs the demo.

- A second machine with the same key and the same tag should reach the same sidebar, the same two tools, and the same stop ids.
- It may not reach the same latency. The free-tier Gemini service has already returned 503 when demand was high, and a 429 when the daily embedding quota was exhausted.
- The embed script resumes after a 429. The chat path now explains a 503 without pasting it.

9.5 What a marker can see without reading the chat code

- STOP_CONDITIONS in orchestrate.py is the list S1 to S8.
- The failure catalogue names the six Week 7 misses and says which five were repaired.
- The README lists the two limits that remain: the 0.70 miss, and turns that can run past 10 seconds.
- A change to a queue name or a blocked phrase should be followed by the deterministic script. That script is the regression check for this release. It is not a substitute for a later live run, and the report does not pretend that it is.

10. Results, Limitations and Professional Reflection

10.1 What works

- A freshman can ask how to activate webmail and see a cited answer when the passage clears 0.70.
- After an explicit click, the student can leave one draft for staff.
- Injection attempts, grade changes, fee decisions, and non-freshman users are refused by code.
- A reader can inspect a turn log and see which stop fired and which tool ran.

10.2 What does not work yet

- Similarity at 0.70 misses some passages a person would judge relevant, especially fee policy and a few joining-instruction details.
- The safe response is a refusal, and that refusal can be wrong from the student's point of view.
- Latency can exceed 10 seconds when the model is slow or returns 503 under load. The student now sees a short retry message, but the answer is still delayed.
- There is no staff inbox, no authentication, and no deletion schedule for drafts. Those are product limits, not hidden features.

10.3 Human responsibility

The model is not the system of record. A support answer that affects a fee, a place, or a case status has to be owned by a member of staff. The agent prepares a draft and shows its evidence. It does not close the case.

Future work, if the university authorised it:

- A staff review screen.
- A retention period for drafts.
- A retrieval evaluation before any change to the 0.70 cutoff. Until that evaluation exists, the cutoff stays.

10.4 Cost and latency

- The project uses the Gemini free tier. There is no separate budget line and no paid deployment.
- A grounded turn is one embedding comparison locally plus one or two generate calls.
- Week 7 saw turns above the 10-second target, up to 13.6 seconds. That is acceptable for a demonstration and poor for a busy helpdesk.
- The response in this release is honesty in the log and a calm sentence in the chat, not a claim that the free tier is a production service.

If the university later hosted this agent, the first work would be a staff queue with names on it, an authentication boundary, a retention rule for drafts, and a fresh retrieval score before anyone moved 0.70.

10.5 What the demonstration should show

- A successful webmail answer shows the cited source.
- A turn log shows the retrieval count, the tool, and the stop id.
- The fee-refund question shows the remaining retrieval miss, or the second-year sentence shows the scope refusal that Week 8 closed.
- The agreement button shows the human gate.

None of those moments require the agent to be trusted on tone. Each one can be checked against a file or a stop id.

References

Huyen, C. (2025). AI Engineering: Building Applications with Foundation Models. O'Reilly Media.

Lanham, M. (2025). AI Agents in Action. Manning Publications.

Osmani, A. (2025). Beyond Vibe Coding: From Coder to AI-Era Developer. O'Reilly Media.

Group M Evening. (2026). Repository evidence cited in the appendix: Agent Task Contract, prompt specification, tool catalogue, corpus source register, memory note, MCP-style interface specification, Week 3 rag_eval.md, Week 7 evaluation report, failure catalogue, and observability samples. Makerere University, BSE4104. https://github.com/SumayyaN/BSE41014_capstone_group_m_eve

Appendix A. Evidence index

This appendix is outside the 8-12 page body. It points at files in tag v1.5.0.

- docs/requirements/Agent Task Contract M EVE.pdf
- docs/requirements/Group_M_User_Stories_and_Acceptance_Criteria.pdf
- docs/requirements/Group M - Prompt Specification v1.0.pdf
- docs/requirements/Tool_Catalogue_and_API_Schemas.pdf
- docs/requirements/Group M - Corpus Source Register.pdf
- docs/requirements/Memory_Design_and_Data_Handling_Note.pdf
- docs/requirements/MCP_Interface_Specification create_support_case.pdf
- docs/Architecture/Group_M_eve_State_Model.pdf
- docs/evaluation/rag_eval.md
- docs/evaluation/week7_evaluation.md
- docs/evaluation/failure_catalogue_week7.md
- docs/evaluation/week8_retest.json
- docs/evaluation/week8_live_results.csv
- docs/evaluation/week8_runner_input.xlsx
- docs/evaluation/Group_M_eve_Observability_Tracing_Evidence.pdf
- docs/evaluation/guardrail_Implementation_Evidence.pdf
- docs/evaluation/Group_M_eve_Final_Evidence_Pack.pdf
- evidence/traces/T1_grounded_answer.json, T2_case_draft.json, T3_failure_recovery.json
- evidence/traces/O1_blocked_topic.json through O4_model_unavailable.json
- src/run_deterministic_checks.py and src/week8_retest.py

Appendix B. Week 8 retest

- Deterministic suite after the fixes: 16 PASS, 0 FAIL.
- The code retest lets B06 and B07 through and still blocks an admissions decision, a grade change, and a second-year sentence.
- The 24 October live run confirms B07 and N03.
- I02 still misses at 0.70. B06 is refused by the model.
- M01 did not finish because the Gemini quota returned 429.

Appendix C. AI Engineering Log

This appendix is outside the 8-12 page body. It declares the tools and models used for retrieval, the running agent, live evaluation, design, coding, testing, and documentation. Each row records the check made before that output was accepted.

Tool or model	Used for	Material output	Check before acceptance	Data sent
gemini-embedding-001	Retrieval	Embeddings for the eight public freshman-onboarding documents. Week 3 index: 90 chunks	Week 3 set, 11 of 15 at a 0.70 cutoff. Compared with 0.50 and 0.65. The cutoff was left at 0.70	Public university documents only. The live timetable site was excluded
gemini-3.6-flash	The running agent, and live evaluation	Answers and tool requests under prompt v1.3. The 24 October retest of 19 live chats	A scenario passes only when task, grounding, tool choice, scope, the 10-second target, and safety all pass. Replies are in week8_live_results.csv. A 503 or 429 is stored in the turn log and shown to the student as "The assistant is temporarily busy. Please try again in a few minutes."	Scenario text only. No ACMIS record, fees ledger, or live student database. The API key stays in .env
Cursor	Design	The decision, written into the report and the code, that prompt text is guidance and the stop conditions are the control. Fee, grade, and admissions decisions are blocked in code before the model is called	Checked against the Agent Task Contract and prompts/v1.3_tools.txt . The prompt still says the assistant must not decide fees or grades. The code enforces it	No secrets and no student records
Cursor	Coding	The five Week 8 fixes in src/tools/case_store.py and src/tools/orchestrate.py: policy questions no longer blocked, fee-	From src/: python run_deterministic_checks.py, 16 of 16, including C07, C08, R04, and R05. python week8_retest.py, 5 of 5. Tagged v1.5.0 after that run	No .env and no drafts.json

Tool or model	Used for	Material output	Check before acceptance	Data sent
		decision language rejected in every category, placeholder and short summaries rejected, second-year through fifth-year phrasing stops at S4, and a 503 or 429 returns a short sentence		
Cursor	Testing	src/week8_retest.py, the runner input rebuilt from the Week 7 results, and the retry rule in src/run_scenarios.py for the busy sentence	The code retest does not need an API key. The live run used docs/evaluation/week 8_runner_input.xlsx. M01 stopped on a 429 quota error and is not scored as a product failure. The Week 7 figure 29/36 was not rewritten	Scenario text and the results CSV
Cursor	Document ation	The final engineering report, the evidence pack, and the Week 8 progress report	Figures checked against week7_evaluation.md, failure_catalogue_wee k7.md, week8_retest.json, week8_live_results.cs v, and the README. References are the three course books only: Huyen (2025), Lanham (2025), and Osmani (2025)	No restricted data